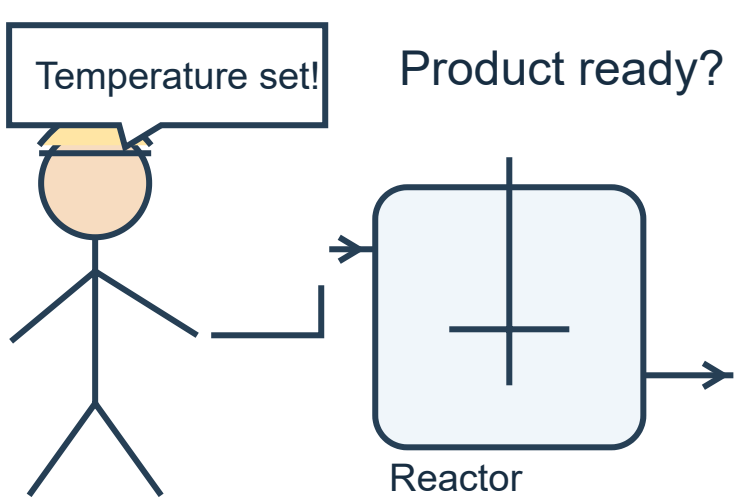
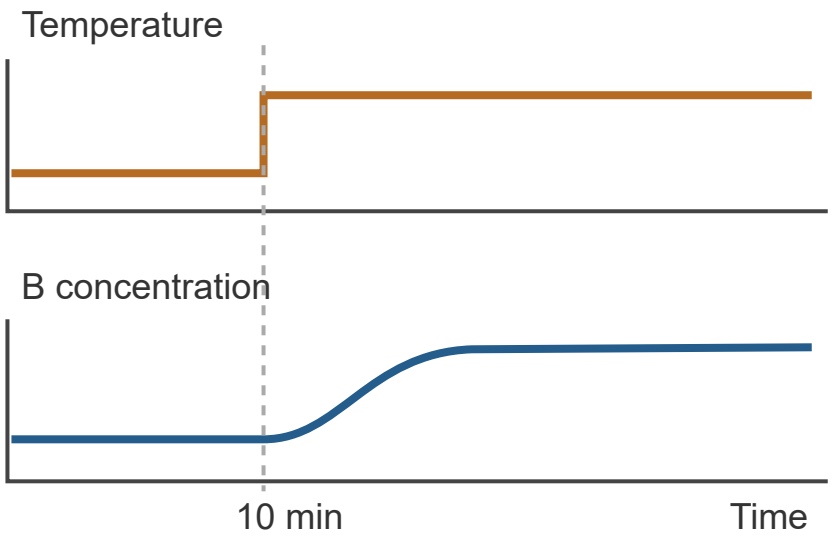
Week 3: Predicting Dynamic Responses and Rollout

The temperature changed. Is the product ready?

A process engineer changes reactor temperature at 10 min. If the outlet flows directly into a product tank, how will B concentration change over the next 20 min?

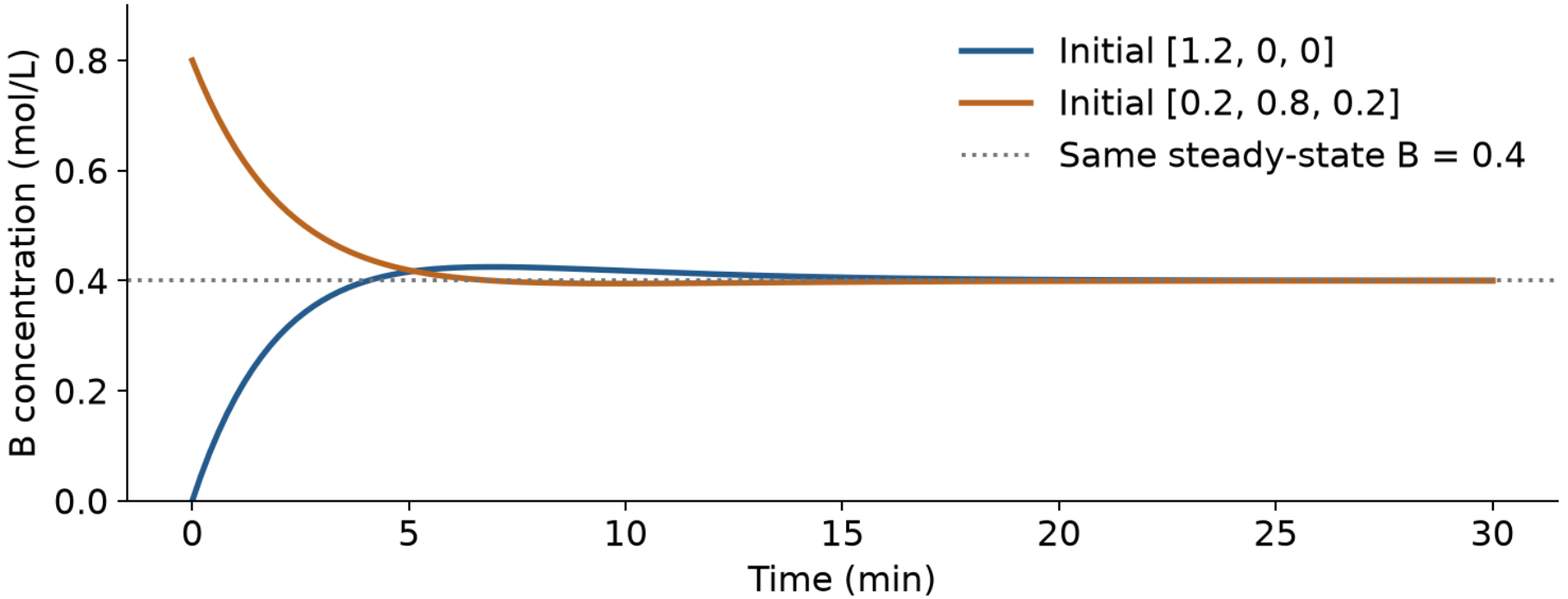




The journey to a new steady state matters

During a grade change or startup, material produced before the target composition is reached can require separation or reprocessing. The endpoint alone does not describe that transient.

At the same temperature, different current tank compositions can lead to different next responses. Learn a short state transition and apply it repeatedly to predict the full trajectory.



T = 330 K, τ = 5 min, C_Af = 1.2 mol/L; only the initial state differs.

What we will build

1. Define the current state, prescribed future inputs, fixed context, and sampling interval.
2. Train a state-transition MLP on concentration increments.
3. Compare one-step predictions and 30 min rollout on held-out trajectories.

Submit the same model’s component errors in both evaluations, error versus horizon, and two temperature-plan trajectories. RNN/LSTM and encoder-decoder are optional extensions.

A steady-state map becomes a transition map

Week 2: [T, τ, C_Af] → the composition after equilibration.

Week 3: current composition + interval input + context → composition one interval later.

$$c_{k+1} = F_{\Delta t}\left(c_k, u_k; q\right)$$

Repeatedly applying this map produces a trajectory. A feedforward MLP can model a dynamic transition when its inputs contain the state needed for that transition.

State, future input, and fixed context

Quantity	Meaning	Unit / role
$c_k = [C_{A,k}, C_{B,k}, C_{C,k}]$	Current reactor composition	mol/L; state
$u_k = T_k$	Temperature over interval k	K; prescribed input
$q = [\tau, C_{Af}]$	Residence time and feed concentration	min, mol/L; fixed per trajectory
$\Delta t = 0.2$	Sampling interval	min; fixed for this model

Specify c_0 and the complete temperature sequence before rollout. The temperature plan is supplied by the exercise.

The reference CSTR

An ideal, perfectly mixed reactor has constant volume and equal inlet/outlet flow. Its consecutive first-order reactions are A → B → C with 1:1 stoichiometry. Feed contains A only. Temperature is prescribed; an energy balance is not included.

$$\frac{dC_A}{dt} = \frac{C_{Af} - C_A}{\tau} - k_1 C_A$$

$$\frac{dC_B}{dt} = k_1 C_A - \frac{C_B}{\tau} - k_2 C_B$$

$$\frac{dC_C}{dt} = k_2 C_B - \frac{C_C}{\tau}$$

$k_1(T) = 0.2 \exp[6000(1/330 - 1/T)]$, $k_2(T) = 0.1 \exp[5000(1/330 - 1/T)]$ in min^{-1}. At 330 K, $k_1 = 0.2$ and $k_2 = 0.1$ min^{-1}.

A continuous process, sampled at fixed times

For each interval [t_k, t_k + Δt), hold T_k constant. Integrate the balances over that interval to obtain c_k+1. The state stays continuous when the input switches.

30 min / 0.2 min = 150 transitions, with 151 sampled states. The temperature step at 10 min starts at interval index 50.

The learned map predicts a 0.2 min transition. Increasing the number of applications extends the horizon; it does not change the sampling interval.

An exact interval response for this teaching model

With constant interval inputs, the balances are affine in the state. Write dc/dt = Mc + b. The steady state $c*$ satisfies $Mc* + b = 0$.

$$F_{\Delta t}\left(c, u; q\right) = c^{*} + e^{M\Delta t}\left(c - c^{*}\right)$$

The lab generates labels at the selected Δt from this analytic solution. Work through the derivations of M and the component formulas in the reading companion.

At c_0 = [1.2, 0, 0], T = 330 K, τ = 5 min, feed = 1.2 mol/L: $c_1 \approx$ [1.153870, 0.045672, 0.000458] mol/L.

Build trajectories before building transition rows

Use 300-360 K, τ = 1-10 min, and feed = 0.8-1.5 mol/L. Random temperature plans contain six 5 min blocks. Each trajectory has its own initial composition, with nonnegative components summing to feed.

Split	Trajectories	Transition rows
Train	60	9000
Validation	15	2250
Test	20	3000

What does one training row contain?

Fields	Construction
Input v_k	[C_A,k, C_B,k, C_C,k, T_k, τ, C_Af]
Next-state label	c_k+1 from the reference interval response
Increment target	Δc_k = c_k+1 − c_k
Identification	split, trajectory ID, t_k

$$\Delta c_k = c_{k+1} - c_k$$

$$\hat{c}_{k+1} = c_k + \mu_\Delta + s_\Delta \odot g_\theta\left(\tilde{v}_k\right)$$

The target is a change in mol/L.

The transition MLP and its scaling

Use $6 \to 32 \to 32 \to 3$, ReLU hidden layers, and a linear output. The parameter count is $(6\times32 + 32) + (32\times32 + 32) + (32\times3 + 3) = 1379$.

Compute input and increment means/standard deviations from train rows. Apply those same statistics to validation, test, and rollout states.

The MLP predicts standardized increments. Restore each increment to mol/L before adding it to the current state. An increment model can represent a zero change, but its architecture alone does not guarantee accuracy or physical consistency.

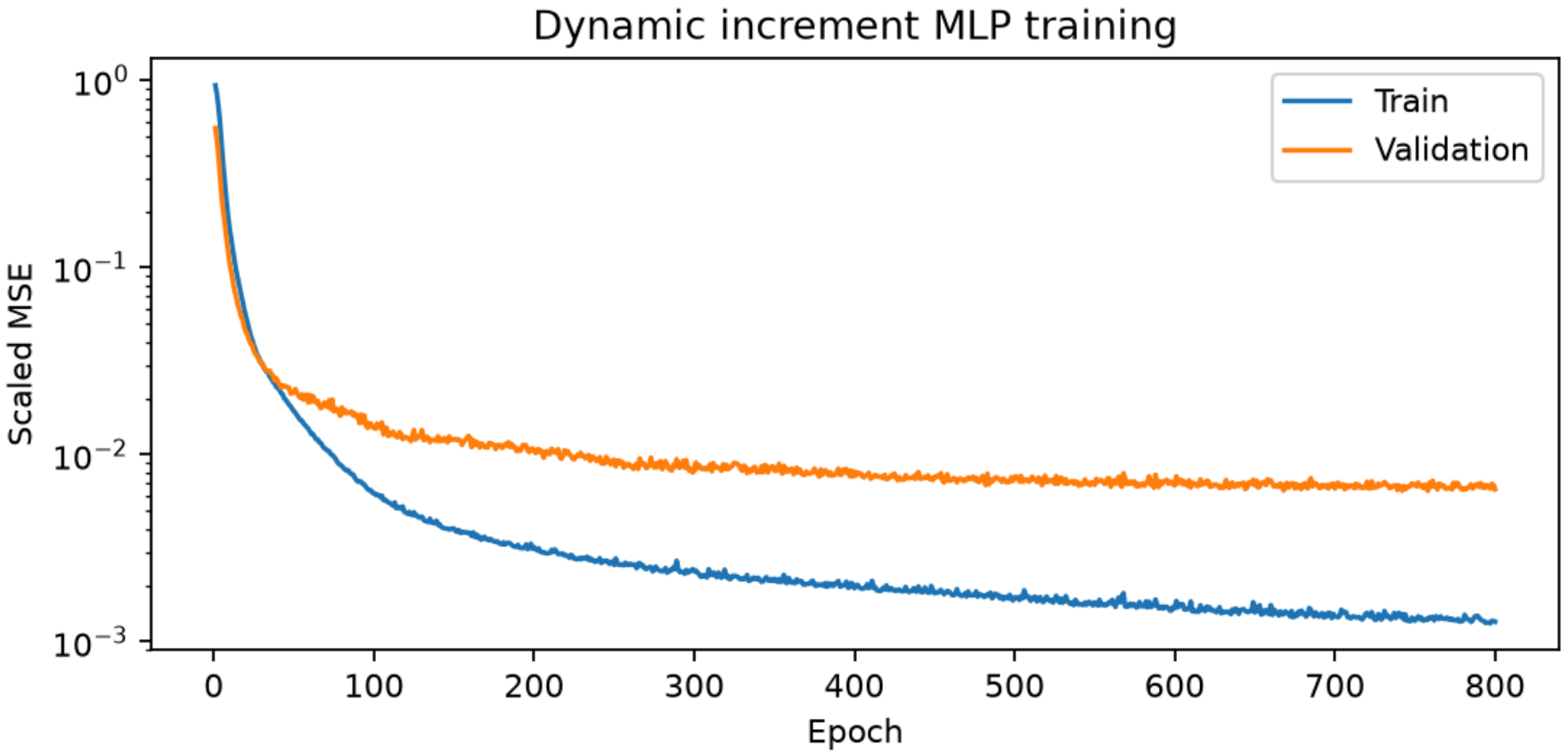
Train on reference current states

Each train input contains the reference current state. Minimize the mean squared standardized increment error.

$$L_1 = \frac{1}{3N} \sum_n \sum_j \left(g_\theta(\tilde{\mathrm{v}}_n)_j - \Delta\tilde{\mathrm{c}}_{n,j} \right)^2$$

Adam: learning rate 0.001, batch size 512, up to 800 epochs. Retain the weights with minimum validation loss. Test trajectories are used after this selection.

Training history



Two ways to evaluate the same model

$$\text{one-step:}\hat{c}_{k+1} = f_\theta\left(c_k, u_k; q\right)$$

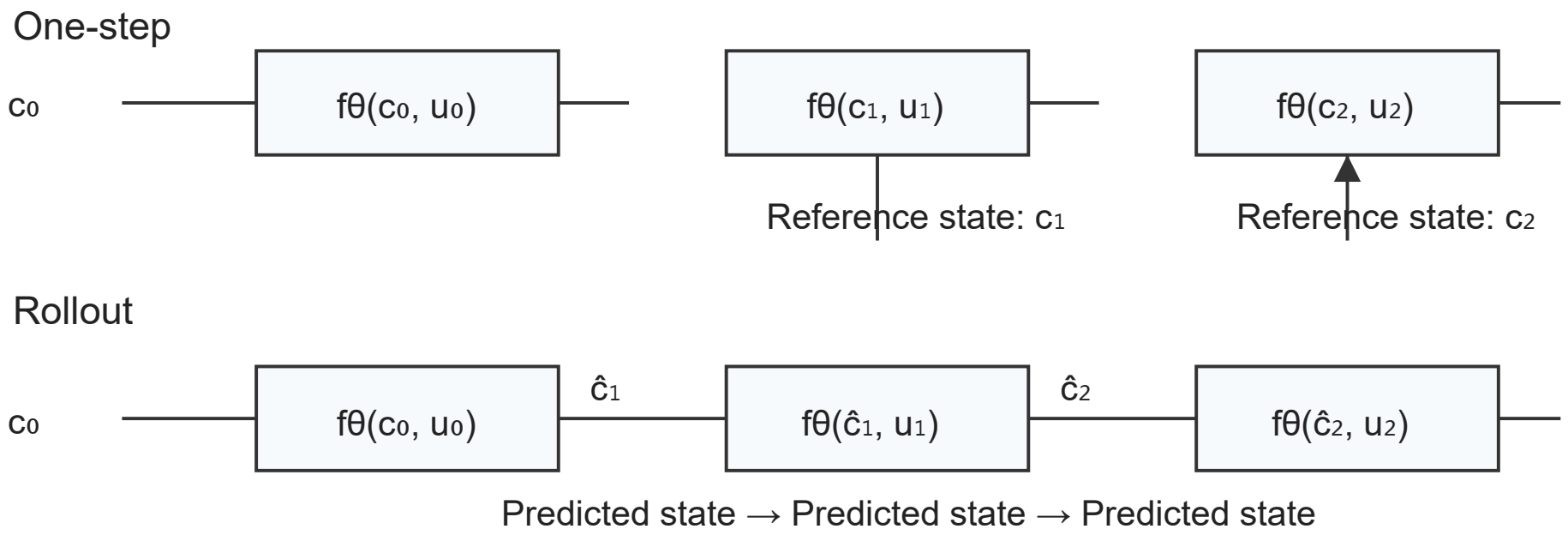
$$\text{rollout:}\hat{c}_{k+1} = f_\theta\left(\hat{c}_k, u_k; q\right), \hat{c}_0 = c_0$$

One-step: reset the input to the reference state at every time.

Rollout: start at c_0, then feed each predicted state into the next transition. Keep all future inputs and the context identical in both evaluations.

One-step evaluation receives a new reference state at every time, so it uses more state information than a forecast initialized only once.

Where the next state comes from



Both diagrams use the same frozen transition model.

A rollout is a short, explicit loop

Given an initial state and 150 future input intervals:

```
pred = np.empty((len(controls) + 1, 3))
pred[0] = initial_state
for k, control in enumerate(controls):
    pred[k + 1] = dynamic_step(
        model, scaling, pred[k:k+1], control[None, :]
    )[0]
```

No reference state enters this loop after initialization. Evaluate all 150 predicted states.

How local error propagates

Let e_k = ĉ_k − c_k. Add and subtract the prediction at the reference state:

$$e_{k+1} = f_\theta\left(\hat{c}_k, u_k\right) - f_\theta\left(c_k, u_k\right) + \delta_k$$
$$\delta_k = f_\theta\left(c_k, u_k\right) - F_{\Delta t}\left(c_k, u_k\right)$$

The first difference transports the existing state error through the learned map. δ_k is the local prediction error at the reference state. Even a small δ_k enters each successive prediction.

Small one-step error can accumulate

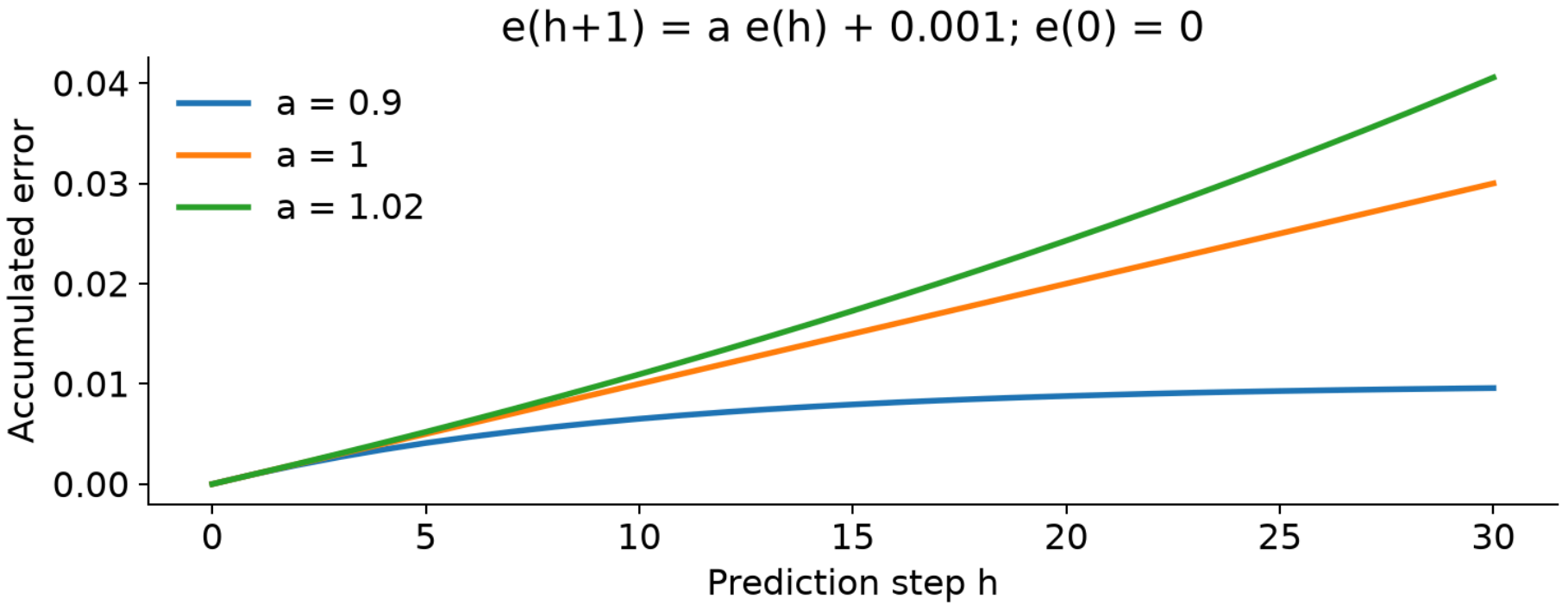
For a scalar illustration, let the reference be F(c) = 0.9c and the learned map be f(c) = 0.9c + 0.001. Start with e_0 = 0.

$$e_{h+1} = 0.9e_h + 0.001$$

$$e_h = 0.001\frac{1 - 0.9^h}{1 - 0.9}$$

One-step bias is 0.001; at 10 recursive steps the error is about 0.00651. A contracting map limits the accumulation. A different state sensitivity can produce a different horizon pattern.

The state sensitivity matters too



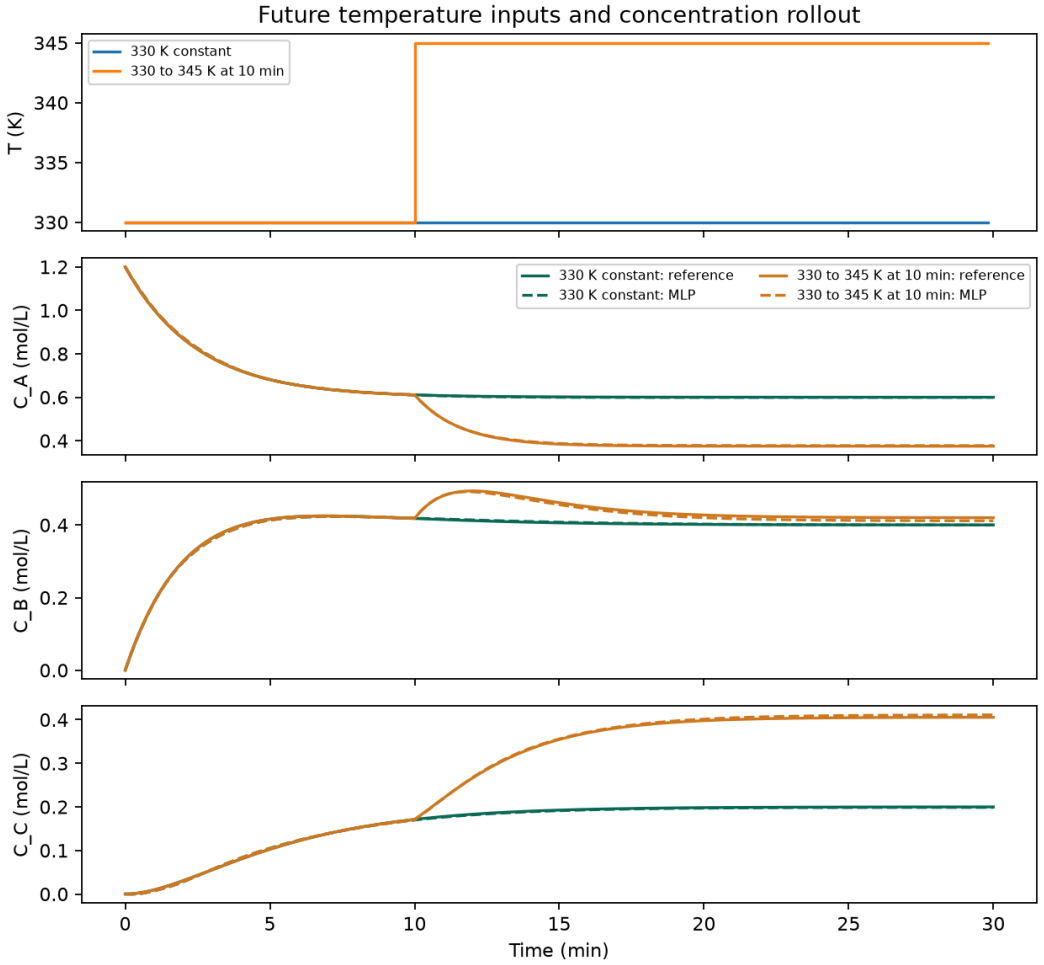
These are scalar examples, not fitted CSTR results. Actual rollout errors can increase, level off, or change sign.

Two prescribed temperature plans

Setting	Constant plan	Step plan
Initial composition, mol/L	[1.2, 0, 0]	[1.2, 0, 0]
τ; C_Af	5 min; 1.2 mol/L	5 min; 1.2 mol/L
$0 \le t < 10$ min	330 K	330 K
$10 \le t \le 30$ min	330 K	345 K

Compare reference trajectories and the MLP rollout under exactly the same plans. Check when the curves separate and how they approach their new steady response.

A, B, and C over the full trajectory



Compare error in physical units

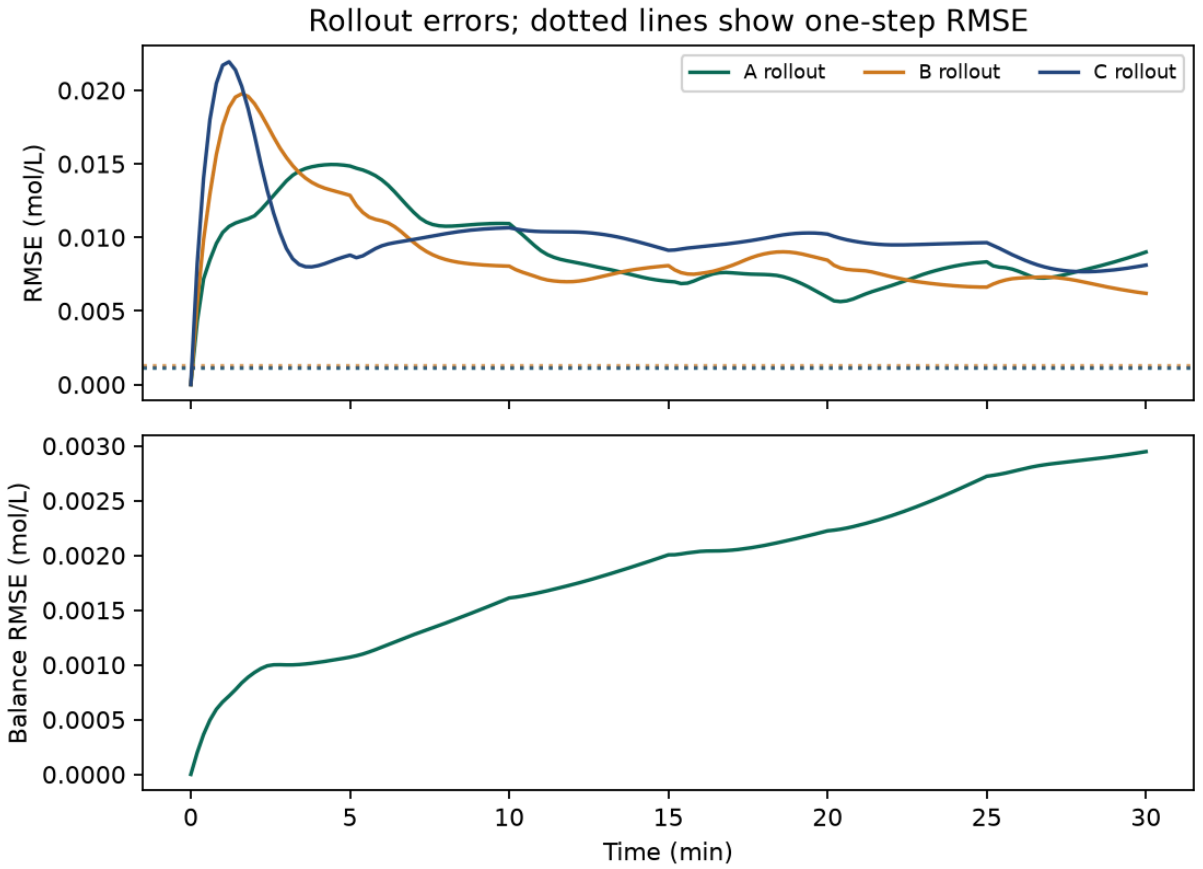
RMSE, mol/L	A	B	C
One-step	0.001170	0.001288	0.001163
Rollout	0.009465	0.009629	0.010425

Results from 20 held-out trajectories, 150 predicted times each. Rollout errors are larger in this stored run. Compute per-component RMSE across the 3000 predictions, then also inspect each horizon separately.

$$\mathrm{RMSE}_{j,h} = \sqrt{\frac{1}{N}\sum_{n}\left(\hat{C}_{n,h,j} - C_{n,h,j}\right)^2}$$

Here n indexes test trajectories, h the number of transitions, and j the component.

Read the error versus horizon



Physical checks along the trajectory

$$S = C_A + C_B + C_C, \frac{dS}{dt} = \frac{C_{\mathrm{Af}} - S}{\tau}$$

$$S_{k+1} = C_{\mathrm{Af}} + \left(S_k - C_{\mathrm{Af}}\right) e^{-\Delta t / \tau}$$

With $S_0 = C_{Af}$ and fixed feed, the reference sum stays equal to feed. The learned outputs need not satisfy this relation.

Check, mol/L	One-step	Rollout
Sum-residual RMSE	0.000051	0.002014
Minimum prediction	-0.001152	-0.005166

Return to the engineer’s question

B at 30 min, mol/L	Reference	MLP rollout
Constant 330 K	0.400091	0.399819
330 → 345 K at 10 min	0.420021	0.411526

Both calculations give more B at 30 min under the step plan. The MLP underestimates that endpoint by about 0.00850 mol/L. The full trajectory reveals the transient that a final steady-state value cannot describe.

A time-dependent product specification would require checking the relevant interval, not just this final sample.

Run the local lab and submit the comparison

The supplied lab bundle includes the reference, stored model, scaling, plots, and notebook.

```
python -m pip install -r requirements.txt
python week03_lab.py --output-dir week03-results
# Optional: train again with the same data construction
python week03_lab.py --train --output-dir week03-retrained
```

Submit state/input definitions, split/scaling details, one-step and rollout RMSE, horizon plots, and the two temperature-plan trajectories.

Exercises and direct reading

1. Derive dS/dt and state the condition for S to remain equal to feed.
2. Compute the first 0.2 min transition and compare it with forward Euler.
3. Explain why the same MLP receives different state inputs in the two evaluations.
4. Change an initial state while keeping the input plan fixed. Compare both reference and predicted trajectories.

Work through the derivations, examples, code, and answer checks in the reading companion. It develops the error recurrence and treats multi-step training as an optional extension.

Optional: when a hidden state helps

The core example supplies all three concentration states and prescribed temperature. If measurements omit relevant states, a history of observations and inputs can help reconstruct them.

$$h_k = \varphi\left(W_x x_k + W_h h_{k-1} + b\right)$$
$$\hat{y}_k = W_y h_k + b_y$$

An RNN updates a learned hidden state. An LSTM adds gated memory updates. Compare these models with the same trajectory split and forecast information.

Implementation: PyTorch RNN and LSTM.

Optional: an encoder and a trajectory decoder

Encoder: summarize past observations and inputs into a state representation.

Decoder: predict the future concentration sequence using that representation and the prescribed future input plan. Ensure each future input reaches the corresponding prediction step.

Read Sutskever, Vinyals & Le (2014), Sections 1-2, for the sequence-to-sequence idea.

Use the original paper for reference and work through the additional derivations in the companion.